

Digital Nurture 5.0 DeepSkilling

Microservices with Spring Boot 2 and Spring Cloud

Tasks:

1. Creating Microservices for account and loan

Account/

Account.java

```
package com.cognizant.account.model;

public class Account {

    private String number;
    private String type;
    private double balance;

    public Account() {
    }

    public Account(String number, String type, double balance) {
        this.number = number;
        this.type = type;
        this.balance = balance;
    }

    public String getNumber() {
        return number;
    }

    public void setNumber(String number) {
        this.number = number;
    }

    public String getType() {
        return type;
    }

    public void setType(String type) {
        this.type = type;
    }

    public double getBalance() {
        return balance;
    }

    public void setBalance(double balance) {
        this.balance = balance;
    }
}
```

AccountController.java

```
package com.cognizant.account.controller;

import org.springframework.web.bind.annotation.*;

import com.cognizant.account.model.Account;

@RestController
public class AccountController {

    @GetMapping("/accounts/{number}")
    public Account getAccount(
        @PathVariable String number) {

        return new Account(
            number,
            "Savings",
            234343
        );
    }
}
```

AccountApplication.java

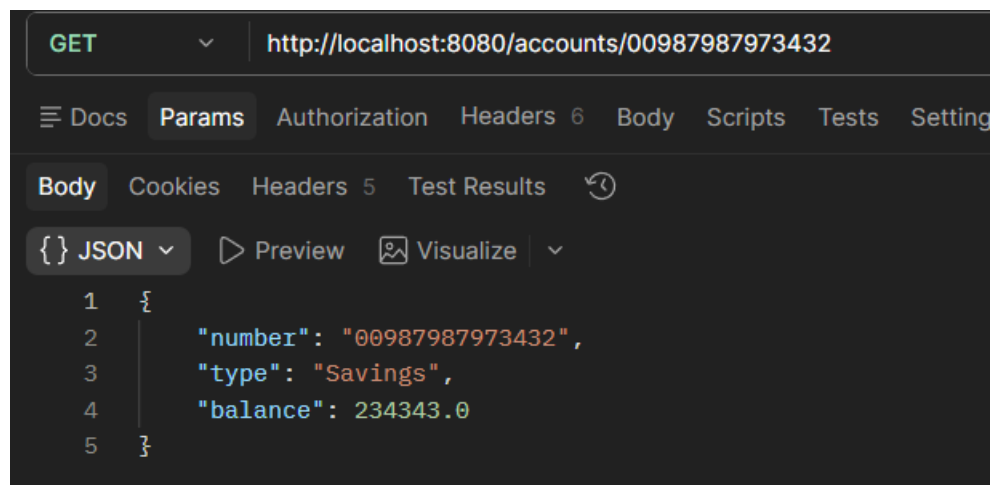
```
package com.cognizant.account;

import org.springframework.boot.SpringApplication;

@SpringBootApplication
public class AccountApplication {

    public static void main(String[] args) {
        SpringApplication.run(AccountApplication.class, args);
    }
}
```

Output



The screenshot shows a web browser interface with a GET request to `http://localhost:8080/accounts/00987987973432`. The response is displayed in JSON format, showing the account details for the specified number.

```
{
  "number": "00987987973432",
  "type": "Savings",
  "balance": 234343.0
}
```

Loan/

Loan.java

```
package com.cognizant.loan.model;

public class Loan {

    private String number;
    private String type;
    private double loan;
    private int emi;
    private int tenure;

    public Loan() {
    }

    public Loan(String number, String type, double loan, int emi, int tenure) {
        this.number = number;
        this.type = type;
        this.loan = loan;
        this.emi = emi;
        this.tenure = tenure;
    }

    public String getNumber() {
        return number;
    }

    public void setNumber(String number) {
        this.number = number;
    }

    public String getType() {
        return type;
    }

    public void setType(String type) {
        this.type = type;
    }
}
```

```
    public double getLoan() {
        return loan;
    }

    public void setLoan(double loan) {
        this.loan = loan;
    }

    public int getEmi() {
        return emi;
    }

    public void setEmi(int emi) {
        this.emi = emi;
    }

    public int getTenure() {
        return tenure;
    }

    public void setTenure(int tenure) {
        this.tenure = tenure;
    }
}
```

LoanController.java

```
package com.cognizant.loan.controller;

import org.springframework.web.bind.annotation.GetMapping;

@RestController
public class LoanController {

    @GetMapping("/loans/{number}")
    public Loan getLoan(@PathVariable String number) {

        return new Loan(
            number,
            "Car",
            400000,
            3258,
            18
        );
    }
}
```

LoanApplication.java

```
package com.cognizant.loan;

import org.springframework.boot.SpringApplication;

@SpringBootApplication
public class LoanApplication {

    public static void main(String[] args) {
        SpringApplication.run(LoanApplication.class, args);
    }

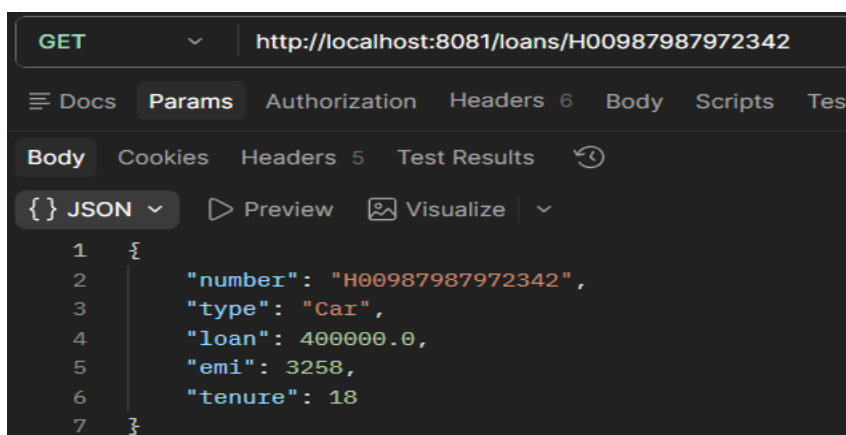
}
```

Application.properties

```
spring.application.name=loan

server.port = 8081
```

Output



The screenshot shows a REST client interface with a GET request to `http://localhost:8081/loans/H00987987972342`. The response is a JSON object with the following fields:

```
{
  "number": "H00987987972342",
  "type": "Car",
  "loan": 400000.0,
  "emi": 3258,
  "tenure": 18
}
```